

DE BEERS GROUP

CONSULTING CHALLENGE

Submitted by:
Trineta Dhara, IITM
S R Janani, IITM



WHY PLASTIC WORKED?

- Lowest cost per unit**
cheapest material + forming cost at scale

Ubiquitous supply chain
Mature supplier network, standardised filling lines, widely available at mass-market volumes
- Lightweight logistics** low dead weight = lower shipping cost per litre

Pressure & barrier durability :
Handles carbonation pressure and prevents contamination across all SKUs

ALTERNATIVES ANALYSIS

Material	Cost	Supply Chain Readiness	Functional Fit	6M Scalability	Sustainability	Still Water	Carbonated Drinks	Juice
rPET	●	●	●	●	●	✓ Primary	✓ Primary	~ Short shelf only
Aluminum	●	●	●	●	●	✓ Premium	✓Premium	✓Secondary
Cartons	●	●	●	●	●	~ Consumer percentration risk	✗	✓ Primary
Paper	●	●	●	●	●	✗	✗	✗
Glass	●	●	●	●	●	~Premium	~Premium	~Premium

SKU MAPPING

CORE NEEDS OF EACH SKU:

Each SKU is governed by a key functional constraint that alternative packaging solutions must meet to be considered viable.

STILL WATER

Lowest cost + high volume distribution.
Highest volume SKU, margin is razor thin:
cost is everything

CARBONATED DRINKS

Pressure resistance + CO₂ barrier
Carbonation is lost without *structural integrity*.

JUICE

Contamination protection + shelf life
Oxidation and microbial risk make barrier properties critical

TRANSITION ECONOMICS

COST OF TRANSITION

Cost Head	rPET	Aluminum	Cartons
Manufacturin g Retooling	Low — existing PET lines adaptable	High — new canning lines needed	Medium — new filling equipment
Supplier Contracts	Low — existing PET supplier base	Medium — new can suppliers	Medium — Tetra Pak/carton suppliers
Compliance & Certification	Low	Medium	Medium
Overall	● Low	● High	● Medium

RECOVERY TIMELINE

Material	Month 0–2	Month 3–4	Month 5–6
rPET	Setup losses	Stabilisation	Break-even
Aluminum	Heavy setup losses	Partial stabilisation	Near break-even
Cartons	Setup losses	Stabilisation	Break-even

FINAL RECOMMENDATION

Tier 1 — Compliance at Scale (Month 0–6)

- Ensure rapid, low-disruption transition
- rPET enables immediate scale for still water and carbonated beverages through existing supply chains and minimal retooling
- Aluminum supports selective premium formats without delaying compliance
- Cartons provide a ready solution for juices, leveraging established supplier ecosystems and shelf-life advantages

Tier 2 — Competitive Moat (Month 6 onwards)

- Defend, differentiate, and scale advantage
- rPET anchors mass-market volume through low cost and EPR compliance, sustaining market share
- Aluminum builds the premium layer with full recyclability, cross-SKU applicability, and pricing power
- Glass protects HoReCa and on-trade channels, preserving margins without impacting mass logistics

LOSSES DURING SWTICHOVER

Loss Source	rPET	Aluminum	Cartons
Production Downtime	Low	High	Medium
Higher Cost/Unit Pre-Scale	Low	Medium	Medium
Distribution Disruption	Low	Medium	Low
Overall	● Low	● High	● Medium

SUMMARY

Material	Transition Cost	Switchover Loss	Recovery Time
rPET	● Low	● Low	3–4 months
Aluminum	● High	● High	5–6 months
Cartons	● Medium	● Medium	3–4 months